GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY
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ESSENTIALS OF ARTIFICIAL INTELLIGENCE

PROGRESS REVIEW REPORT

CareerSense AI

An Explainable AI-Powered Career Development and Skill-Roadmap Platform for Undergraduate Students

Project Details	
Project Title	CareerSense AI — An Explainable AI-Powered Career Development and Skill-Roadmap Platform for Undergraduate Students
Module	Essentials of Artificial Intelligence
Group No.	Group 19

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Content

1.Abstract 3
2.Introduction 4
3.Problem background 4
3.1. Problem Statement 5
4.Objectives 5
4.1.Primary objective 5
4.2. Specific Objectives 5
4.3.Target Users 6
5.Selected AI techniques 6
6. Justification of AI techniques 7
6.1. K-Nearest Neighbors and Decision Tree 7
6.2. Rule-Based Expert System 8
6.3.A* Search 8
6.4. Why a Hybrid Pipeline 8
7.System Design and Architecture 9
7.1 End-to-End Student Workflow 9
7.2. Data Design 9
7.3. Technology Stack 10
8.Implementation details 11
8.1. Layer-1 Machine-Learning Classification 11
8.2. Layer 2 – Rule-Based Expert System 12
8.3.Layer 3 – A* Roadmap Optimization 13
8.4.Platform and Sample Outputs 13
8.5.Security 16
9. Testing and Results 16
9.1 Evaluation Protocol 16
9.2 Model Results 17
9.3 Automated Software Testing 18
10.Discussion, Limitations and Ethics 19
10.1. Key Findings 19
10.2. Limitations 19
10.3. Ethical and Social Considerations 19
11.Conclusion 20
12.References 20
13.Contribution of Group Members 21

1. Abstract

While there is an increasing number of career paths that are available to undergraduate computing students – Software Engineering, Data Science / AI, Cybersecurity, Cloud / DevOps and UI/UX Design – the general and often vague and scarce advice they receive is rarely aligned with their transcript, skills and study time. This report introduces CareerSense AI, an explainable decision-support and skill-roadmap solution for the Faculty of Computing, General Sir John Kotelawala Defence University (KDU).

The platform combines 3 complementary techniques within an AI pipeline: (1) machine learning classification (KNN as key algorithm, DT as baseline interpretable algorithm); (2) a forward-chaining Rule Based Expert System to validate academic prerequisites and skill dependencies, prioritise skill gaps to compute a 4 component Career Readiness Score; (3) an A* search over the learning-activity graph that generates a prerequisite-respecting roadmap scheduled against the student's weekly study hour budget. Implementing this system in Python using Streamlit, Scikit-learn and SQLite, it offers Student, Academic Advisor, Programme Coordinator and Administrative dashboards.

The K-NN model achieved an accuracy of 88.82% (weighted F1 88.87%) when trained on a stratified 170-record held-out test set on a synthetic cohort of 850 student profiles, and 89.56% ± 2.48% in five-fold cross-validation, compared to a baseline accuracy of 60.00% achieved by the Decision Tree model. There are 74 unit and integration tests, and they all pass automatically. In the report, there is also a dataset-realism problem that was identified during the progress review where it was found that the data was almost perfect because the synthetic data was separable, and how this issue was addressed.

Keywords: *career guidance, explainable AI, K-Nearest Neighbors, rule-based expert system*, *A* search,skill-gap analysis, Streamlit.*

2. Introduction

However, most students select a career based on the advice of peers and/or the expectations that family members have for them, or on a general knowledge source available online, rather than on an analytical review of their own strengths. Counselling units at the university level are usually small, and one counsellor won't be able to read the files on each student's grades, skills and interests in detail. The impact is that there is a mismatch between students and specializations and skill gaps are identified only near graduation.

The gap was addressed by creating the CareerSense AI for the module Essentials of Artificial Intelligence. The key design principle is that career advice should be explainable: a recommendation is only useful if the student and the advisor can understand why it was made, what is missing from the student's profile and what concrete, time realistic steps will make up the gap. Therefore, the system is not based on a single black box model. Instead, it combines statistical prediction, explicit expert rules and heuristic search – all of which solve different aspects of the advisory problem.

3. Problem background

Undergraduate computing education covers fields such as Software Engineering, Computer Science, Information Technology, Information Systems, Computer Engineering and Data Science. Undergraduate computing education also splits into career paths. Each path requires skills and has its own prerequisites. Students must pick electives, project subjects, certifications and internships early. Many students do this without an idea of how their current grades and skills fit what employers want.

Today students mostly get advice from three places. First the occasional one-on-one advice from staff is hampered by students per advisor so meetings are brief and rare. Second, web career pages and online roadmaps show job titles and long lists of tools, but they do not look at a student’s own grades, skill levels or how busy a semester is. Third, friends and family give help, but that help is usually not based on facts.

The results are clear. Students spend time on tools before they have the basics. Students pick specializations that do not suit their strengths. Students find big skill gaps near graduation. These problems can bring down grades, lower motivation and lead to careers. Undergraduate computing

education needs a cheap tool that looks at a student's grades, skills and interests and gives clear doable next steps.

3.1. Problem Statement

Students do not yet have access to a system that provides them with up to date data on their academic performance, skills and interests, identifies career pathways that suit them and identifies the specific gaps in their skills, and transforms these gaps into an actionable plan that is tailored to the time available for them to study. There is currently a lot of existing guidance which is manual, generic and is not suited to individual cases.

4. Objectives

4.1.Primary objective

To create a system that uses intelligence to give personalized career advice and help with development. This system will look at profiles of students studying computer science at the level. It will give recommendations about possible career paths. It will also check for missing skills. Show where students need to improve. The system will give a score to show how ready students are and create the learning plans for them.

4.2 Specific Objectives

- Analyze students' academic records, domain interest and students' self-rate technical skills by applying AI techniques.
- Predict and rank top 5 career tracks for a person w.r.t. his profile with a confidence score attached to each of top 5 ranks.
- Ensure that academic prerequisites and skill dependencies are validated using clear and editable rules from experts and report prioritised skill gaps.
- Calculate an AIR understandable Career Readiness Score and Tier based on competency, academic, project and documentation evidence.

- Produce an optimized learning roadmap with respect to the student's study hours per week, while adhering to the prerequisites.
- Provide feedback tools and anonymous cohort analysis to support advisors and programme coordinators.
- Create an accessible, many-faceted web site and an individualized career guidance document that can be downloaded.

4.3.Target Users

It is used by undergraduate computing students (primary users), Academic Advisors who get AI summaries to help guide mentoring sessions, Programme Coordinators who require anonymized, cohort-level information for curriculum planning, and System Administrators who keep up the knowledge bases and review the performances of the models.

5. Selected AI techniques

The final system is composed of three techniques of AI, one for each stage in the pipeline (Table 1). The guiding principle of the Stage 1 proposal was the development of an Artificial Neural Network, content-based similarity matching and rule-based reasoning, which were refined during the development process to better align with the explainability goal, as outlined in Table 2.

6. AI layer	Technique	Role in the system
Layer 1	K-Nearest Neighbors (k = 7, distance-weighted) – primary; Decision Tree (max depth 6) – interpretable baseline	Ranks the five career tracks for a student profile; probabilities blended 70% K-NN / 30% Decision Tree; K-NN neighbour distances and tree feature importances support explanation
Layer 2	Rule-Based Expert System (forward chaining, 13 rules)	Validates academic prerequisites and skill dependencies, prioritises skill gaps and computes the Career Readiness Score and tier

6. AI layer	Technique	Role in the system
Layer 3	A* search over a 22-activity learning graph	Finds a minimum-effort, prerequisite-respecting sequence of learning activities and schedules it into weekly milestones

Table 1: Selected AI techniques and their roles.

Aspect	Proposal (Stage 1)	Final system
Classifier	Artificial Neural Network	K-NN primary model plus Decision Tree baseline, blended 70% / 30%
Ranking	Content-based similarity matching of job roles	Ranked match across five career tracks with K-NN peer-similarity evidence
Rule-based reasoning	Filtering and validation of recommendations	Forward-chaining expert system with a four-component readiness score
Improvement planning	List of skill gaps and suggestions	A* search producing a time-budgeted weekly roadmap
Users / interface	Students and counsellors; Streamlit or Flask	Student, Advisor, Coordinator and Administrator portals in Streamlit with SQLite

6. Justification of AI techniques

6.1. K-Nearest Neighbors and Decision Tree

K-NN was selected as the key classifier since the suitability to a career is naturally a similarity: a student will be best fit for the careers of his peers whose grades, interests and skills are most similar. The method is non-parametric, does not require special treatment of multiple classes, is helpful for a relatively small tabular dataset (850 records) and produces neighbour distances that the student can use as evidence. The "explain" goal would be violated as a neural network, as originally proposed, would make learning more opaque but not more accurate on this small dataset.

The Decision Tree is intentionally kept as a "white box" baseline model. Less accurate, but the splits and importances of the Gini indicate which grades, interests or skills are important to a classification, and blending it at 30% ensures that an interpretable component is still in the pipeline without overwhelming the powerful K-NN prediction.

6.2. Rule-Based Expert System

Machine learning can detect correlations, but not to ensure hard requirements are met – a student with an inadequate Programming grade should not be told that they are instantly eligible for advanced Software Engineering courses. Explicit rules are deterministic checks (which is always auditable), that an advisor can read and question. The rules and competency benchmarks can be stored in JSON files, enabling lecturers to easily update the rules as industry needs change, without altering application code.

6.3.A* Search

Skill gaps to plan is a path finding problem with activities having costs (study hours) and prerequisites. A* is well suited because it will be able to produce a plan with minimum costs and dependency consistently always, and it will explore less states than uninformed search. A simple list of resources would not account for ordering and the time constraint of the student; A* with weekly scheduling would take both into account.

6.4. Why a Hybrid Pipeline

No single technique can be entirely reliable; each method is a complement to its weaknesses: K-NN estimates gaps which are based on a track's profile but are not a description of gaps; the rules are a description of gaps but cannot rank tracks; A* plans the remediation but requires gaps as input. They work together from "What suits me?" to "What am I missing?" to "What should I do? What order? When?".

7. System Design and Architecture

CareerSense AI has a layered design. Role-specific Streamlit portals interface with an application/service layer which handles session authentication, profile and document management, report generation and cohort aggregation. Underneath it is the three AI layers, with a common data layer comprised of the synthetic dataset, three JSON knowledge files and a SQLite database. The personalized advisory report is rendered by a reporting layer.

7.1 End-to-End Student Workflow

A student registers (or logs in as a Quick Demo Access student), selects a career track, sets weekly study hours and logs module results for six subject areas. The student then self-rates 25 technical competencies on a four-level scale (None, Beginner, Intermediate, Advanced), uploads documents (CV, transcript, certificates) and documents project evidence. The AI Career Assessment tab displays the ranked list of track matches and competency radar, while the rule engine generates the skill gaps and also the readiness score, and the A* optimiser creates the weekly roadmap, whose milestones the student checks off as Planned / In Progress / Completed – completing one will update the skill level and re-calculate the readiness score. Advisors provide feedback statements and action items – student can download formatted Career Advisory Report.

7.2. Data Design

In accordance with the proposal's dataset strategy, there was no real institutional dataset of student profiles and career outcomes, so a synthetic dataset was created consisting of 850 records. It represents the KDU Faculty of Computing students in 6 degree programmes and 4 years of study. Its properties are shown in Table 3, where 36 out of the 43 columns represent the model input features. Class distribution is Software Engineering 251 (29.5%), Cloud / DevOps 178 (20.9%), Data Science / AI 165 (19.4%), Cybersecurity 148 (17.4%) and UI/UX Design 108 (12.7%).

Attribute group	Columns	Type / scale
Identifiers / metadata	student_id, name, reg_no, degree, year	Identifier

Attribute group	Columns	Type / scale
Academic performance	gpa; grade_programming, grade_math, grade_database, grade_networking, grade_systems, grade_design	Numeric, 0.0–4.0 grade points
Domain interests (5)	interest_software, _data, _security, _cloud, _design	Ordinal, 1–5
Technical skills (25)	e.g. skill_oop, skill_dsa, skill_ml, skill_docker, skill_figma	Ordinal, 0–3 (None to Advanced)
Target label	career_track	Categorical, five classes

Table 3: Dataset attributes.

Persistent data (users, profiles, grades, skills, projects, documents and roadmaps) is stored in SQLite while the three knowledge bases (career_definitions.json, rules_knowledge_base.json and learning_graph.json) are separated from code and can be maintained separately.

7.3. Technology Stack

Component	Details
Language / UI	Python 3; Streamlit web interface; Plotly competency radar charts
AI / ML	Scikit-learn (KNeighborsClassifier, DecisionTreeClassifier, StandardScaler), NumPy, Pandas; custom rule engine and A* optimiser
Persistence	SQLite with parameterised queries; trained models stored as .pkl files

Component	Details
Modules	ai_engine/ (ml_classifier, rule_engine, a_star_roadmap, explainability); modules/ (auth, student, advisor, coordinator and admin views, report generator); database/; scripts/; tests/
Quality	74 automated unit/integration tests and a GitHub Actions workflow that runs them on each push

Table 4: Technology stack and code organisation.

8. Implementation details

The prototype is a working Streamlit application. The Student portal has tabs for AI Career Assessment, A* Learning Roadmap, Competencies & Skills Academic Records, Projects & Portfolio Documents & CV Advisor Feedback and Download Report. The Advisor portal supports the Advisor search for advisees, review of AI explanations and prerequisite audits and feedback notes; the Coordinator portal shows the Coordinator portal anonymous interest distributions and skill-deficiency heatmaps; and the Administrator portal provides the Administrator portal model explorer with metrics, confusion matrices, rule-base inspection and model retraining.

8.1. Layer-1 Machine-Learning Classification

Every student is described by a 36-number vector that contains six grade elements, five interest scores and twenty-five skill levels. These numbers are normalised with StandardScaler, which is fitted on the training data before being fed to K-NN because K-NN relies on distance and needs consistent scales. The Decision Tree on the hand uses the raw numbers directly. K-NN looks at seven neighbours and weighs them by distance while the Decision Tree is limited to a depth of six. The class probabilities from K-NN and Decision Tree are blended by giving seventy percent weight to K-NN and thirty percent weight to Decision Tree to produce the ML competency score for each track. Once training is complete the models are saved as knn_model.pkl, dt_model.pkl and scaler.pkl so that the application can load them without needing to retrain. The score shown to students is an advisory affinity index, not a calibrated probability. It balances across the five tracks. Mixes the ML competency score, with domain interests, a degree-alignment prior and if the student declares a target aspiration, that aspiration as well. The weights adapt to how complete the profile's for a fully complete profile the weights might be forty percent ML, thirty percent

target, fifteen percent interests and fifteen percent degree prior. The raw ML prediction is kept separate so that it is never influenced by the student's stated preference.

8.2. Layer 2 – Rule-Based Expert System

The expert system (ai_engine/rule_engine.py) contains thirteen rules in three categories:

• Academic prerequisite rules (five): grade in a foundational subject for each track for example Programming B- (2.7) for Software Engineering; Mathematics for Data Science; Networking for Cybersecurity; Operating Systems for Cloud / DevOps; HCI/Design for UI/UX.

• Competency dependency rules (five): for example Automated Testing requires at least Beginner Object-Programming; CI/CD requires Docker; Machine Learning requires the Python data stack.

• Portfolio and readiness rules (three): flags for no recorded project, no CV and a weekly study budget that is unrealistic for the number of open gaps.

Inference is forward-chaining. Each prerequisite is marked Pass, Fail or Missing; each required competency's compared with the student's level to give a gap of zero to three levels and a priority of Satisfied, Medium or High where High applies to gaps of two or more levels or, to core-priority competencies; dependency violations are flagged.

The Career Readiness Score is the sum of four components mapped to a tier (Table 5).

Component	Weight	Readiness tier (total score)
Competency alignment	50%	≥ 80%: Job / Internship Ready
Academic prerequisites	25%	60–79%: Near Ready (Minor Gaps)
Project evidence	15%	40–59%: Developing (Targeted Training Needed)
Documentation / CV completeness	10%	< 40%: Foundational Stage (Significant Gaps)

Table 5: Career Readiness Score components and tiers.

8.3.Layer 3 – A* Roadmap Optimization

The learning graph contains twenty-two activities, which include courses, projects and practice tasks. Seven activities belong to Software Engineering, five to Data Science / AI, four to Cybersecurity three to Cloud / DevOps and three to UI/UX Design. Every activity lists its estimated hours the skill it targets the level it grants and the activities that must be finished first.

• State: the skill levels and the set of activities that have already been completed.

• Transitions: an activity can be started after all of its prerequisite activities are finished.

• Path cost g(n): the total of estimated study hours that have been accumulated.

• Heuristic h(n): a lower bound on the effort still needed. It is made by picking, for each skill that is still missing the activity that can close that gap. If the skill gaps have candidate activities the cheapest costs are added together. If one activity can raise skills at once the bound uses the highest of the cheapest costs for each skill – this is called the bottleneck bound. This bound never overestimates, so A* remains admissible.

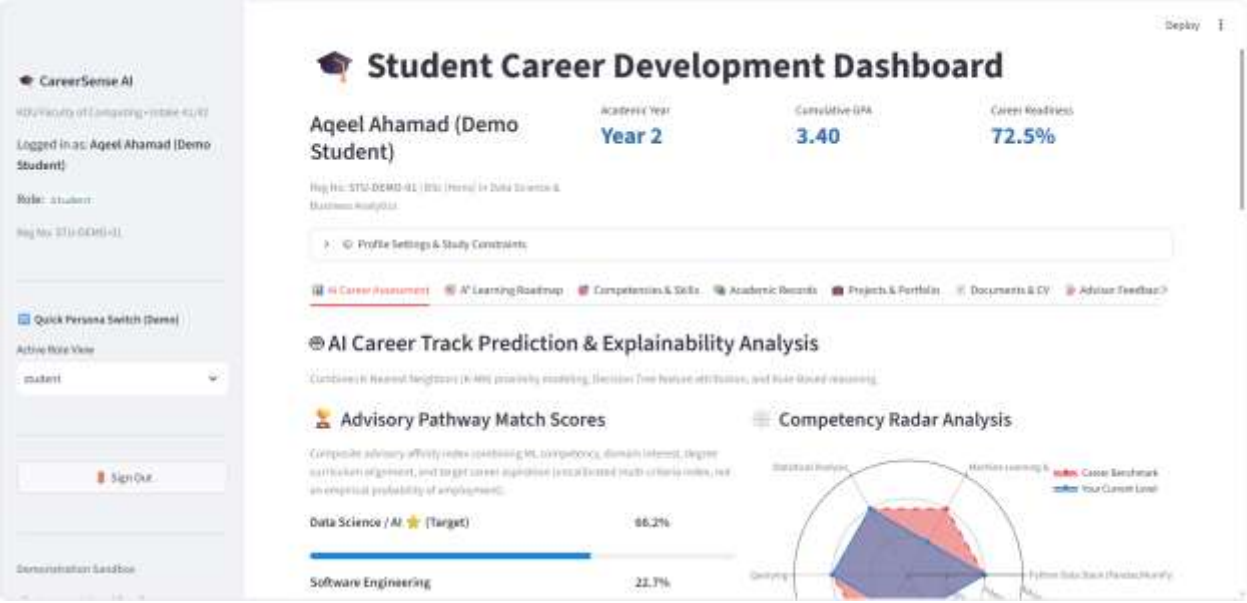
• Goal test: all required competencies of the target track have reached their level.

The search is limited to two thousand iterations. If the learning graph cannot fill every skill gap or the iteration limit is reached a deterministic topological-order fallback produces a plan. That plan is marked as not optimal and is shown with the status partial_coverage, goal_unreachable or search_limit_reached. Only plans that come from a verified A* search are labelled optimal. The final sequence of activities is then broken into milestones that fit the students declared hours each week. If an activity needs time than one week it is spread over several weeks and no week gets more hours than its limit.

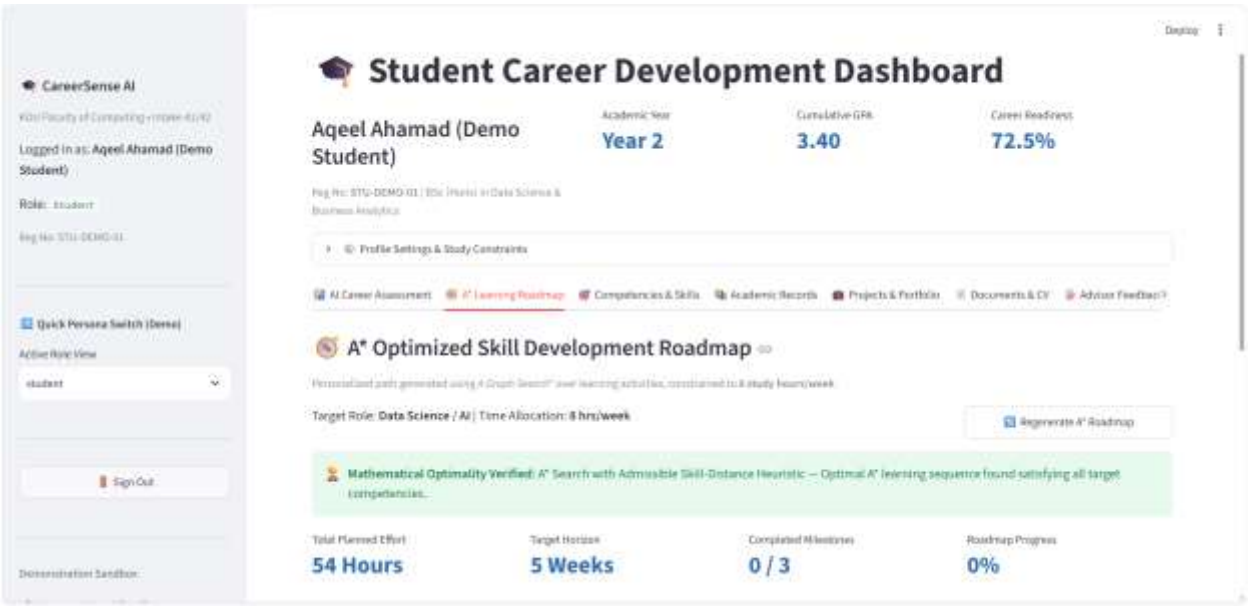
8.4.Platform and Sample Outputs

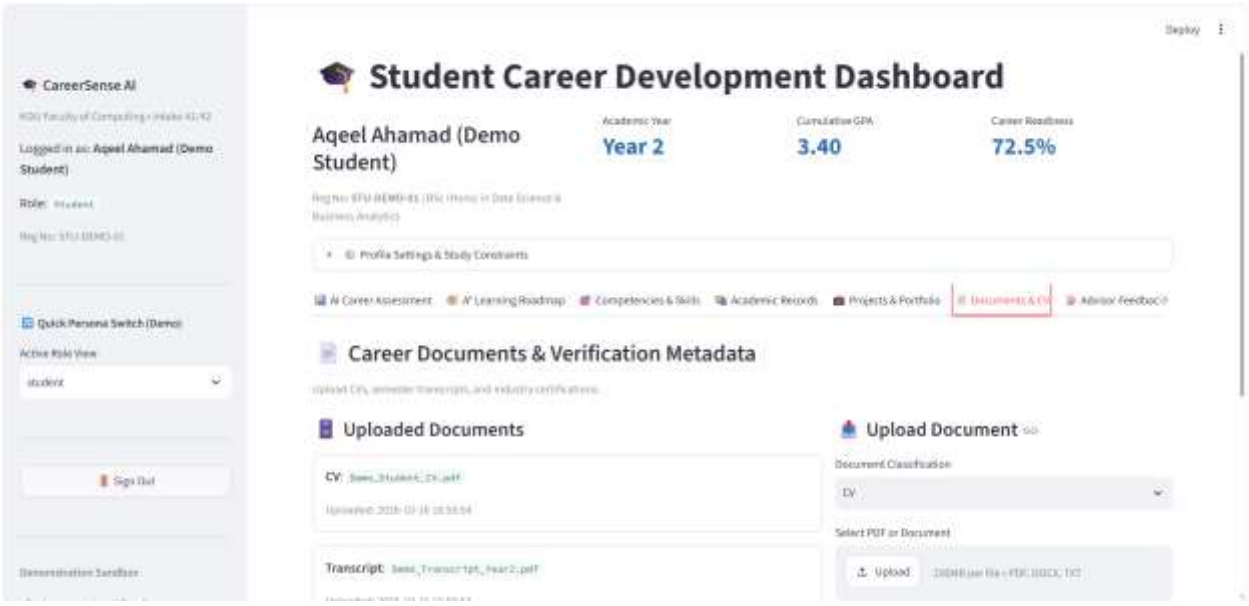
Streamlit application that offers four portals. The Student portal has tabs for AI Career Assessment, A* Learning Roadmap, Competencies & Skills Academic Records, Projects & Portfolio Documents & CV Advisor Feedback and Download Report. The Advisor portal lets users search for advisees look at AI explanations and prerequisite audits and write feedback notes. The Coordinator portal shows interest distributions and heatmaps that reveal skill deficiencies. The

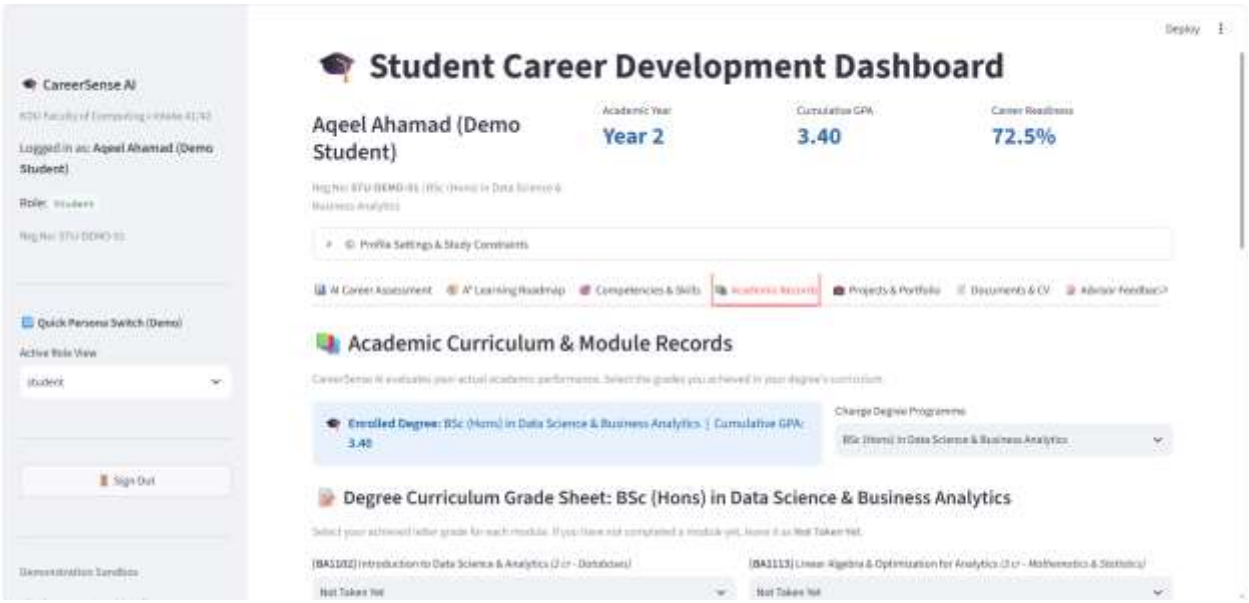
Administrator portal provides a model explorer, with metrics, confusion matrices, rule-base inspection and retraining tools.



For the demonstration student (Year 2 Data Science & Business Analytics target Data Science / AI 8 study hours per week) the rule engine reported a readiness score of 87.9% (tier: Job / Internship Ready) with a breakdown of competency alignment 40.9/50 prerequisites 22.0/25 project evidence 15.0/15 and documentation 10.0/10. The highest-priority gap was Machine Learning & Modeling (Beginner to Intermediate) followed by Data Visualization. The A* roadmap planned 76 hours over a 10-week horizon in five milestones: Pandas & NumPy (12 h) Exploratory Data Analysis & Visualization project (14 h) Applied Statistical Analysis (10 h) Supervised & Unsupervised Machine Learning (18 h) and an End-, to-End Predictive ML project (22 h).


CareerSense AI
Logged in as: Aqeel Ahamad (Demo Student)
Role:
Quick Persona Switch (Demo)
Active Role View
student
Sign Out
Student Career Development Dashboard
Aqeel Ahamad (Demo Student)
Academic Year
Year 2
Cumulative GPA
3.40
Career Readiness
72.5%
Profile Settings & Study Constraints
AI Career Assessment
A* Learning Roadmap
Competencies & Skills
Academic Records
Projects & Portfolio
Documents & CV
Advisor Feedback
A* Optimized Skill Development Roadmap
Target Role: Data Science / AI | Time Allocation: 8 hrs/week
Regenerate A* Roadmap
Mathematical Optimality Verified: A* Search with Admissible Skill-Distance Heuristic — Optimal A* learning sequence found satisfying all target competencies.
Total Planned Effort
54 Hours
Target Horizon
5 Weeks
Completed Milestones
0 / 3
Roadmap Progress
0%


CareerSense AI
Logged in as: Aqeel Ahamad (Demo Student)
Role:
Quick Persona Switch (Demo)
Active Role View
student
Sign Out
Student Career Development Dashboard
Aqeel Ahamad (Demo Student)
Academic Year
Year 2
Cumulative GPA
3.40
Career Readiness
72.5%
Profile Settings & Study Constraints
AI Career Assessment
A* Learning Roadmap
Competencies & Skills
Academic Records
Projects & Portfolio
Documents & CV
Advisor Feedback
Career Documents & Verification Metadata
Uploaded Documents
CV:
Transcript:
Upload Document
Document Classification
CV
Select PDF or Document
Upload



8.5.Security

Access is controlled by role-based authorisation. Passwords are stored with salted PBKDF2-HMAC-SHA256 (100,000 iterations 16-byte salt) and verified with constant-time comparison with smooth migration of legacy hashes at login. Database access uses queries, uploaded file names are sanitised against directory traversal, file types are whitelisted and student assets are isolated by user ID. Production use would additionally require HTTPS/TLS, external secrets management and institutional single sign-, on.

9. Testing and Results

9.1 Evaluation Protocol

The data was randomly split into 80/20 stratified hold-out (680 in training, 170 in test, random_state = 42) for the 850 records. Five-fold stratified cross validation was performed on the training set alone and a scikit-learn Pipeline with StandardScaler fitted within each fold to prevent scaling information from leaking from the folds used for validation into the folds used for testing. Only the set held out was used in the final comparison, and not for tuning.

9.2 Model Results

Model	Role	CV accuracy	Test accuracy	Macro F1
Zero-Rule (majority class)	Sanity baseline	29.56%	29.41%	9.09%
Decision Tree (depth 6)	Interpretable baseline	63.53%	60.00%	59.72%
Random Forest (100 trees)	Ensemble benchmark	87.21%	82.94%	83.13%
Logistic Regression (L2)	Linear benchmark	88.53%	88.82%	89.45%
K-NN (k=7, distance)	**Primary model**	**89.56%**	**88.82%**	**89.49%**

Table 6: Model comparison (held-out test set N = 170).

K-NN got 88.82% accuracy on the test set. Its weighted precision was 89.74% weighted recall was 88.82%. Weighted F1 score was 88.87%. When we did cross-validation the average accuracy was 89.56% with a deviation of 2.48%. The individual fold scores were 91.18%, 88.97%, 85.29%, 92.65% and 89.71%.

Logistic Regression had the test accuracy as K-NN.. We picked K-NN because it uses peer-based reasoning and gives clear distance-based evidence that's easy to check.

The Decision Tree model only reached 60.00% accuracy. The important features, for it were interest_security (0.148) interest_design (0.122) interest_software (0.111) interest_data (0.109) and interest_cloud (0.096). After those came skill_dsa skill_web_api and skill_ml.

K-NN class	Prec.	Recall	F1	Support	Correct	Main confusion
Software Engineering	79.66	94.00	86.24	50	47	–
Data Science / AI	93.55	87.88	90.62	33	29	4 → SE
Cybersecurity	95.83	76.67	85.19	30	23	4 → SE, 3 → Cloud

K-NN class	Prec.	Recall	F1	Support	Correct	Main confusion
Cloud / DevOps	88.89	91.43	90.14	35	32	2 → SE
UI/UX Design	100.00	90.91	95.24	22	20	2 → SE

Table 7: K-NN per-class results (%) on the held-out test set.

Errors appear mainly in fields. Cybersecurity students are often confused with Software Engineering and Cloud / DevOps. Cybersecurity and Software Engineering share secure-coding, Linux and networking skills. This overlap is natural. Shows that the benchmark cannot be split easily.

9.3 Automated Software Testing

The test suite (tests/test_all.py) contains seventy-four tests. Passes in full (re-run while preparing this report). The test suite covers:

• The test suite checks invariants of the evaluation metrics. The test suite checks that confusion-matrix totals match expectations, that row sums class support and that zero division is handled safely.

• The test suit verifies A* correctness. The test suite verifies that A* is optimal when compared to exhaustive/Dijkstra search on controlled graphs. The test suite checks that A* remains admissible with activities that zero-gap cases are handled, that weekly-hour caps are respected that fallback flagging occurs correctly and that scheduled hours equal the accumulated path cost.

• The test suite evaluates rule-engine behavior. The test suite evaluates how the rule-engine behaves for students who meet all benchmarks and for students who meet none. The test suite evaluates that the rule-engine keeps the ML prediction separate, from student aspirations.

• The test suite tests security and data handling. The test suite tests salted hashing legacy-hash migration, SQL-injection resistance, upload sanitization, session purging and multi-user asset isolation.

10. Discussion, Limitations and Ethics

10.1. Key Findings

The synergy of the three paradigms was more helpful than any of the three techniques alone. The classifier provides a “data-driven” perspective on which tracks are similar to the student's profile, the rules transform the perspective into statements that can then be verified as having been met, and A* translates the gaps into a process the student can follow to meet the statements within a specified time budget. Storing the knowledge in the JSON files and maintaining the Decision Tree as a baseline transparency helped achieve the maintainability and explainability objectives of the proposal.

10.2. Limitations

Synthetic data: this is the 850-record dataset; it is a model of plausible structure and not actual student or labour-market data and therefore accuracy statements are indicative of the pipeline, rather than actual performance.

Outputs: Self-reported skills may be over or underestimated; the readiness score partially corrects this with grades, projects and documents.

There are a few activities in Software Engineering, Data Science / AI and UI/UX Design and Cloud/ DevOps, which have a rich graph, so A* has some options there and it often uses the flagged fallback; the graph of Cybersecurity is too sparse and A* has very limited options and may use the flagged fallback there.

Shallow document analysis: CVs and transcripts uploaded are stored with metadata but are not processed with OCR or NLP.

The match percentages (not probabilities) on the track are uncalibrated scores.

10.3. Ethical and Social Considerations

CareerSense AI is a decision support tool, not a decision maker: it offers predictions in the form of affinities ranked by likelihood, with explanations provided; it is the student and advisor's decision to choose a career. Cohort analytics are anonymized and no individual is profiled, access

is restricted to specific roles and protected attributes like gender are not used as features of the model. For a deployment that involved real student data, institutional approval, informed consent, a data-protection review, and a bias audit on real outcomes would be required.

11. Conclusion

CareerSense AI shows how classification and expert-system reasoning and heuristic search can be combined into a single, explainable career-advisory system for undergraduate computer students. It is implemented through 4 role-based portals and passed by 74 automated tests, and it provides the following components: the career tracks are classified by the K-NN algorithm (accuracy on the synthetic test is 88.82%), prerequisites are validated with a 13-rule expert system, gaps are prioritized in order to plan time-budgeted learning, and a time-budgeted learning plan is generated with an admissible A* search. The project also highlighted the value of analyzing unexpected high accuracy: the result of extremely high accuracy at the progress review directly resulted in a more accurate dataset and a more stringent assessment. The final remaining task is to validate the system by using real student data, turning it into an actual advisory tool for students.

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13. Contribution of Group Members

Member	Contribution
MFA Ahamad (D/DBA/25/0021)	Core AI architecture and full-stack lead: designed the three-layer architecture; implemented the K-NN + Decision Tree classifier, rule-based expert system and A* optimiser; built the Streamlit multi-role platform; wrote and ran the automated test suite; trained and evaluated the models; produced project documentation.
RMBP Rathnayake (D/BCS/25/0007)	Assisted with gathering and refining the proposal; reviewed competency definitions used for career tracking in the rule engine; supported testing of the Programme Coordinator dashboard.
SGT Tharumila (D/BIS/24/0032)	Checked and verified the prerequisite and skill-dependency rules against real-world scenarios; supported test-case design for the rule engine and validation of roadmap outputs across student profiles and documentation review.
KKGAM Bandara (D/BIT/24/0041)	Assisted with dataset review; verified sample readiness and roadmap outputs; helped compile the report, presentation slides and demonstration video.